

Hemisphere

A plane passing through the centre cuts the sphere in two equal parts, each part is called a hemisphere.

Let the radius of hemisphere be r , then



- (i) Volume of hemisphere = $\frac{2}{3}\pi r^3$ cu units
- (ii) Curved surface area of hemisphere = $2\pi r^2$ sq units
- (iii) Total surface area = $2\pi r^2 + \pi r^2 = 3\pi r^2$ sq units
- (iv) Volume of hemispherical shell = $\frac{2}{3}\pi(R^3 - r^3)$

where, R = outer radius, r = inner radius

EXAMPLE 19. The volumes of two hemispheres are in the ratio 8:27. What is the ratio of their radii?

- a. 2 : 3 b. 3 : 2 c. 1 : 2 d. 2 : 1

Sol. a. Let volumes be V_1 and V_2 .

$$\therefore V_1 : V_2 = 8 : 27 \Rightarrow \frac{2}{3}\pi r_1^3 : \frac{2}{3}\pi r_2^3 = 8 : 27$$

$$\Rightarrow r_1^3 : r_2^3 = 8 : 27 \Rightarrow r_1 : r_2 = 2 : 3$$

IMPORTANT FORMULAE

- If the side of a cube is increased by $x\%$, then its volume is increased by $\left[\left(1 + \frac{x}{100}\right)^3 - 1\right] \times 100\%$.
- If the length, breadth and height of cuboid are made x , y and z times respectively, then its volume is increased by $(xyz - 1) \times 100\%$.
- If the length, breadth and height of a cuboid are increased by $x\%$, $y\%$ and $z\%$ respectively, then its volume is increased by $\left[x + y + z + \frac{xy + yz + zx}{100} + \frac{xyz}{(100)^2}\right] \%$.
- If the sides and diagonal of a cuboid are given, then the total surface area in terms of diagonal and sides is given by
Total surface area = $(\text{Sum of the sides})^2 - (\text{Diagonal})^2$.
- If the side of a cube is increased by $x\%$, then the surface area is increased by $\left(2x + \frac{x^2}{100}\right) \%$.

> PRACTICE EXERCISE

- Three equal cubes are placed adjacently in a row. The ratio of total surface area of the new cuboid to that of the sum of the surface areas of the three cubes is
(a) 3 : 1 (b) 6 : 5 (c) 7 : 9 (d) 6 : 7
- If side of a cube is increased by 12%, by how much percent does its volume increase?
(a) 35.2% (b) 40.5% (c) 45.0% (d) 42.4%
- The curved surface area of a cylinder is 1320 cm^2 and its base has diameter 21 cm, then the height of the cylinder is
(a) 10 cm (b) 20 cm (c) 22 cm (d) 25 cm
- A cylindrical vessel can hold 154 g of water. If the radius of its base is 3.5 cm and 1 cm^3 of water weights 1 g. The depth of the water is
(a) 2 cm (b) 3 cm
(c) 4 cm (d) 5 cm
- The curved surface area of a cylindrical pillar is 264 m^2 and its volume is 924 m^3 . The diameter of the pillar is
(a) 3 m (b) 6 m
(c) 7 m (d) 14 m
- How many metres of cloth 50 m wide will be required to make a conical tent, the radius of whose base is 7 m and whose height is 24 m?
(a) 9 m (b) 11 m (c) 12 m (d) 13 m
- The radius and height of a right circular cone are in the ratio of 5 : 12 and its volume is 2512 cm^3 . The slant height of the cone is
(a) 24 cm (b) 25 cm (c) 26 cm (d) 27 cm
- If the height of a cone is doubled, then its volume is increased by
(a) 100% (b) 200% (c) 300% (d) 400%
- If the surface areas of two spheres are in the ratio of 4 : 25, then the ratio of their volumes is
(a) 2 : 25 (b) 4 : 75 (c) 8 : 125 (d) 16 : 125
- A cone and a cylinder are of the same height. Their radii of the bases are in ratio of 2 : 1. The ratio of their volumes is
(a) 2 : 1 (b) 3 : 2 (c) 4 : 3 (d) 1 : 3
- If the height and diameter of a right circular cylinder are 32 cm and 6 cm respectively, then the radius of the sphere whose volume is equal to the volume of the cylinder is
(a) 3 cm (b) 4 cm (c) 6 cm (d) 8 cm

- 12.** Two solid spheres of gold having diameters 3 cm and 4 cm are molten and then cast into one big sphere of gold. If the radius of this sphere is x , then what is the value of x^3 ?
 (a) 125 cm^3 (b) 15.625 cm^3
 (c) 11.375 cm^3 (d) 9.875 cm^3
- 13.** The total surface area of a cone, whose slant height is equal to the radius R of its base, is S . If A is the area of a circle of radius $2R$, then which one of the following is correct?
 (a) $A = S$ (b) $A = 2S$ (c) $A = S/2$ (d) $A = 4S$
- 14.** A cylinder having base of circumference 60 cm is rolling without sliding at a rate of 5 rounds per second. How much distance will the cylinder roll in 5 s?
 (a) 15 m (b) 1.5 m (c) 30 m (d) 3 m
- 15.** 27 drops of water form a big drop of water. If the radius of each smaller drop is 0.2 cm, then what is the radius of the bigger drop?
 (a) 0.4 cm (b) 0.6 cm (c) 0.8 cm (d) 1.0 cm
- 16.** A circus tent is made of canvas and is in the form of a right circular cylinder and a right circular cone above it, the diameter and height of the cylindrical part of the tent are 126 m and 5 m, respectively. The total height of the tent is 21 m. Then, the cost of the canvas used for tent at the rate of ₹ 12 per m^2
 (a) ₹ 14850 (b) ₹ 168200 (c) ₹ 178200 (d) ₹ 112000
- 17.** A solid sphere of radius 6 cm is melted into a hollow cylinder of uniform thickness. If the external radius of the base of the cylinder is 5 cm and its height is 32 cm. The uniform thickness of the cylinder is
 (a) 1.5 cm (b) 3 cm (c) 1.2 cm (d) 1 cm
- 18.** Given a solid cylinder of radius 10 cm and length 1000 cm a cylindrical hole is made into it to obtain a cylindrical shell of uniform thickness and having volume equal to one-fourth of the original volume of the original cylinder. The thickness of the cylindrical shell is
 (a) $5(\sqrt{5} - 2) \text{ cm}$ (b) $7(2 - \sqrt{3}) \text{ cm}$
 (c) 10 cm (d) $5\sqrt{2} \text{ cm}$
- 19.** A tent is of the shape of right circular cylinder upto a height of 3 m and then becomes a right circular cone with a maximum height of 13.5 m above the ground. The cost of painting the inner side of the tent at the rate of ₹ 2 per m^2 , if the radius of the base is 14 m is
 (a) ₹ 2048 (b) ₹ 2068 (c) ₹ 2008 (d) ₹ 2088
- 20.** A container is in the form of a right circular cylinder surmounted by a hemisphere of the same radius 15 cm as the cylinder. If the volume of the container is $32400 \pi \text{ cm}^3$, then the height h of the container satisfies which one of the following ?
 (a) $135 \text{ cm} < h < 150 \text{ cm}$ (b) $140 \text{ cm} < h < 147 \text{ cm}$
 (c) $145 \text{ cm} < h < 148 \text{ cm}$ (d) $139 \text{ cm} < h < 145 \text{ cm}$
- 21.** A conical flask of base radius r and height h is full of milk. The milk is now poured into a cylindrical flask of radius $2r$. What is the height to which the milk will rise in the flask?
 (a) $\frac{h}{3}$ (b) $\frac{h}{6}$ (c) $\frac{h}{9}$ (d) $\frac{h}{12}$
- 22.** The radius and height of a right circular cone are in the ratio 3 : 4 and its volume is $96 \pi \text{ cm}^3$. What is the lateral surface area?
 (a) $24 \pi \text{ cm}^2$ (b) $36 \pi \text{ cm}^2$ (c) $48 \pi \text{ cm}^2$ (d) $60 \pi \text{ cm}^2$
- 23.** What is the number of wax balls, each of radius 1 cm, that can be molded out of a sphere of radius 8 cm?
 (a) 256 (b) 512 (c) 768 (d) 1024
- 24.** 10 cylindrical pillars of a building have to be painted. The diameter of each pillar is 70 cm and the height is 4 m. What is the cost of painting at the rate of ₹ 5 per m^2 ?
 (a) ₹ 400 (b) ₹ 440 (c) ₹ 480 (d) ₹ 500
- 25.** The radii of the circular ends of a bucket of height 40 cm are of lengths 35 cm and 14 cm. What is the volume of the bucket?
 (a) 60060 cm^3 (b) 70040 cm^3
 (c) 80080 cm^3 (d) 80160 cm^3
- 26.** If S is the total surface area of a cube and V is its volume, then which one of the following is correct?
 (a) $V^3 = 216 S^2$ (b) $S^3 = 216 V^2$
 (c) $S^3 = 6 V^2$ (d) $S^2 = 36 V^3$
- 27.** A cylindrical tank 7 m in diameter, contains water to a depth of 4 m. What is the total area of wetted surface?
 (a) 110.5 m^2 (b) 126.5 m^2
 (c) 131.5 m^2 (d) 136.5 m^2
- 28.** The radii of two cylinders are in the ratio 2 : 3 and their curved surface areas are in the ratio 5 : 3. What is the ratio of their volumes?
 (a) 20 : 27 (b) 10 : 9 (c) 9 : 10 (d) 27 : 20
- 29.** A cylindrical vessel of height 10 cm has base radius 60 cm. If d is the diameter of a spherical vessel of equal volume, then what is the value of d ?
 (a) 30 cm (b) 60 cm (c) 90 cm (d) 120 cm
- 30.** A hollow sphere of internal and external diameters 4 cm and 8 cm, respectively is melted into a cone of base diameter 8 cm. The height of the cone is
 (a) 11 cm (b) 12 cm (c) 14 cm (d) 16 cm

- 31.** If the diameter of a sphere is doubled, how does its surface area change?
 (a) It increases two times (b) It increases three times
 (c) It increases four times (d) It increases eight times
- 32.** A cistern 6 m long and 4 m wide contains water to a depth of 1.25 m. What is the area of wetted surface?
 (a) 40 m^2 (b) 45 m^2 (c) 49 m^2 (d) 73 m^2
- 33.** The outer and inner diameters of a circular pipe are 6 cm and 4 cm, respectively. If its length is 10 cm, then what is the total surface area in cm^2 ?
 (a) 35π (b) 110π
 (c) 150π (d) None of these
- 34.** A toy is in the form of a cone mounted on a hemisphere such that the diameter of the base of the cone is equal to that of the hemisphere. If the diameter of the base of the cone is 6 cm and its height is 4 cm, what is the surface area in cm^2 of the toy? (Take $\pi = 3.14$)
 (a) 93.62 (b) 103.62 (c) 113.62 (d) 115.50
- 35.** A solid cylinder of height 9 m has its curved surface area equal to one-third of the total surface area. What is the radius of the base?
 (a) 9 m (b) 18 m (c) 27 m (d) 30 m
- 36.** The volume of a sphere is 8 times that of another sphere. What is the ratio of their surface areas?
 (a) 8:1 (b) 4:1 (c) 2:1 (d) 4:3
- 37.** The dimensions of a field are 12 m $\times$ 10 m. A pit 5 m long, 4 m wide and 2 m deep is dug in one corner of the field and the Earth removed has been evenly spread over the remaining area of the field. The level of the field is raised by
 (a) 30 cm (b) 35 cm (c) 38 cm (d) 40 cm
- 38.** A cube of 9 cm edge is immersed completely in a rectangular vessel containing water. If the dimensions of base are 15 cm and 12 cm. Then, the rise in water level in the vessel is
 (a) 4.05 cm (b) 4 cm (c) 3.5 cm (d) 3 cm
- 39.** From a solid cube of edge 3 m, a solid of largest sphere is curved out. What is the volume of solid left?
 (a) $(27 - 2.25\pi) \text{ m}^3$ (b) $(27 - 4.5\pi) \text{ m}^3$
 (c) $2.25\pi \text{ m}^3$ (d) $4.5\pi \text{ m}^3$
- 40.** A rectangular tank is $80 \times 40 \text{ cm}^3$. Water flows into it through a pipe of cross-section area 40 cm^2 at the speed of 10 km/h. The rise in the level of water in the tank in $\frac{1}{2}$ h is
 (a) $\frac{3}{2} \text{ cm}$ (b) $\frac{4}{3} \text{ cm}$ (c) $\frac{5}{8} \text{ cm}$ (d) 6 cm
- 41.** A measuring jar of internal diameter 10 cm is partially filled with water. Four equal spherical balls of diameter 2 cm each are dropped in it and they sink down in the water completely. The change in the level of water in the jar is
 (a) $\frac{16}{65} \text{ cm}$ (b) $\frac{15}{16} \text{ cm}$ (c) $\frac{16}{75} \text{ cm}$ (d) None of these
- 42.** The height of a cone is 30 cm. A small cone is cut off at the top by a plane parallel to the base. If its volume be $\frac{1}{27}$ of the volume of the given cone, then the height above the base where the section is made, is
 (a) 12 cm (b) 15 cm (c) 20 cm (d) 22 cm
- 43.** From a wooden cylindrical block, whose diameter is equal to its height, a sphere of maximum possible volume is carved out. What is the ratio of the volume of the utilised wood to that of the wasted wood?
 (a) 2:1 (b) 1:2 (c) 2:3 (d) 3:2
- 44.** The base diameter of a right circular cylinder is 3 cm. There is a section making an angle of 30° with the cross section. What is its area?
 (a) $\frac{9\pi}{4} \text{ cm}^2$ (b) $\frac{3\sqrt{3}\pi}{2} \text{ cm}^2$ (c) $\frac{9\pi}{8} \text{ cm}^2$ (d) $\frac{9\sqrt{3}\pi}{8} \text{ cm}^2$
- 45.** A cone is inscribed in a hemisphere such that their bases are common. If C is the volume of the cone and H that of the hemisphere, then what is the value of $C:H$?
 (a) 1:2 (b) 2:3 (c) 3:4 (d) 4:5
- 46.** If the diameter of a wire is decreased by 10%, by how much per cent (approximately) will the length be increased to keep the volume constant?
 (a) 5% (b) 17% (c) 20% (d) 23%
- 47.** The diameter of a solid metallic right circular cylinder is equal to its height. After cutting out the largest possible solid sphere S from this cylinder, the remaining material is recast to form a solid sphere S_1 . What is the ratio of the radius of sphere S to that of sphere S_1 ?
 (a) $1:2^{\frac{1}{3}}$ (b) $2^{\frac{1}{3}}:1$ (c) $2^{\frac{1}{3}}:3^{\frac{1}{3}}$ (d) $3^{\frac{1}{3}}:2^{\frac{1}{2}}$
- 48.** A square has its side equal to the radius of a sphere. The square revolves round a side to generate a surface of total area S . If A is the surface area of the sphere, then which one of the following is correct?
 (a) $A = 3S$ (b) $A = 2S$ (c) $A = S$ (d) $A < S$
- 49.** A swimming pool is 24 m long and 15 m broad. When x number of men dive into the pool, the height of the water rises by 1 cm. If the average amount of water displaced by one man is 0.1 m^3 , then what is the value of x ?
 (a) 36 (b) 72 (c) 108 (d) 360

- 50.** Water is distributed to a town of 50000 inhabitants from a rectangular reservoir consisting of 3 equal compartments. Each compartment has length and breadth 200 m, 100 m, respectively and 12 m depth of water in the beginning. The allowance is 20 L per head per day. For how many days will the supply of water hold out?
(a) 240 days (b) 720 days (c) 800 days (d) 900 days
- 51.** A right circular cylinder and a right circular cone have equal bases and equal volumes. But the lateral surface area of the right circular cone is $15/8$ times the lateral surface area of the right circular cylinder. What is the ratio of radius to height of the cylinder?
(a) 3 : 4 (b) 9 : 4 (c) 15 : 8 (d) 8 : 15
- 52.** The volume of a cuboid whose sides are in the ratio of 1 : 2 : 4 is same as that of a cube. What is the ratio of length of diagonal of cuboid to that of cube?
(a) $\sqrt{1.25}$ (b) $\sqrt{1.75}$ (c) $\sqrt{2}$ (d) $\sqrt{3.5}$
- 53.** A field is 125 m long and 15 m wide. A tank $10\text{ m} \times 7.5\text{ m} \times 6\text{ m}$ was dug in it and the Earth, thus dug out was spread equally on the remaining field. The level of the field thus raised is equal to which one of the following?
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm
- 54.** If C_1 is a right circular cone with base radius r_1 cm and height h_1 cm and C_2 is a right circular cylinder with base radius r_2 cm and height h_2 cm and if $r_1 : r_2 = 1 : n$ (where n is a positive integer) and their volumes are equal, then which one of the following is correct?
(a) $h_1 = 3nh_2$ (b) $h_1 = 3n^2h_2$
(c) $h_1 = 3h_2$ (d) $h_1 = n^2h_2$
- 55.** A right circular cone is cut by a plane parallel to its base in such a way that the slant heights of the original and the smaller cone thus obtained are in the ratio 2 : 1. If V_1 and V_2 are respectively the volumes of the original cone and of the new cone, then what is $V_1 : V_2$?
(a) 2 : 1 (b) 3 : 1 (c) 4 : 1 (d) 8 : 1
- 56.** The surface area of a sphere is 616 cm^2 . If its radius is changed so that the area gets reduced by 75%, then the radius becomes
(a) 1.6 cm (b) 2.3 cm (c) 2.5 cm (d) 3.5 cm
- 57.** A sphere is inscribed in a cubical box such that the sphere is tangent to all six faces of the box. What is the ratio of the volume of the cubical box to the volume of sphere?
(a) 6π (b) 36π (c) $\frac{4\pi}{3}$ (d) $\frac{6}{\pi}$
- 58.** From a solid cylinder of height 4 cm and radius 3 cm, a conical cavity of height 4 cm and of base radius 3 cm is hollowed out. What is the total surface area of the remaining solid?
(a) $15\pi\text{ cm}^2$ (b) $22\pi\text{ cm}^2$ (c) $33\pi\text{ cm}^2$ (d) $48\pi\text{ cm}^2$
- 59.** The curved surface of a cylinder is 1000 cm^2 . A wire of diameter 5 mm is wound around it, so as to cover it completely. What is the length of the wire used?
(a) 22 m (b) 20 m
(c) 18 m (d) None of these
- 60.** In order to fix an electric pole along a roadside, a pit with dimensions $50\text{ cm} \times 50\text{ cm}$ is dug with the help of a spade. The pit is prepared by removing Earth by 250 strokes of spade. If one stroke of spade removes 500 cm^3 of Earth, then what is the depth of the pit?
(a) 2 m (b) 1 m (c) 0.75 m (d) 0.5 m
- 61.** A figure is formed by revolving a rectangular sheet of dimensions $7\text{ cm} \times 4\text{ cm}$ about its length. What is the volume of the figure thus formed?
(a) 352 cm^3 (b) 296 cm^3
(c) 176 cm^3 (d) 616 cm^3
- 62.** The diagonals of the three faces of a cuboid are x, y, z , respectively. What is the volume of the cuboid?
(a) $\frac{xyz}{2\sqrt{2}}$
(b) $\frac{\sqrt{(y^2 + z^2)(z^2 + x^2)(x^2 + y^2)}}{2\sqrt{2}}$
(c) $\frac{\sqrt{(y^2 + z^2 - x^2)(z^2 + x^2 - y^2)(x^2 + y^2 - z^2)}}{2\sqrt{2}}$
(d) None of the above
- 63.** The volume of a cone is equal to that of a sphere. If the diameter of base of cone is equal to the diameter of the sphere, what is the ratio of height of cone to the diameter of the sphere?
(a) 2 : 1 (b) 1 : 2 (c) 3 : 1 (d) 4 : 1
- 64.** The length, breadth and height of a rectangular parallelopiped are in ratio 6 : 3 : 1. If the surface area of a cube is equal to the surface area of this parallelopiped, then what is the ratio of the volume of the cube to the volume of the parallelopiped?
(a) 1 : 1 (b) 5 : 4 (c) 7 : 5 (d) 3 : 2
- 65.** A hollow hemisphere is made of a sheet of a metal 1 cm thick. If the outer radius is 5 cm. What is the weight of the hemisphere (1 cm^3 of the metal weight 9 g)?
(a) $54\pi\text{ g}$ (b) $366\pi\text{ g}$ (c) $122\pi\text{ g}$ (d) $108\pi\text{ g}$

- 66.** Half of a large cylindrical tank open at the top is filled with water and identical heavy spherical balls are to be dropped into the tank without spilling water out. If the radius and the height of the tank are equal and each is four times the radius of a ball, what is the maximum number of balls that can be dropped?

(a) 12 (b) 24 (c) 36 (d) 48

- 67.** A cylinder is circumscribed about a hemisphere and a cone is inscribed in the cylinder so as to have its vertex at the centre of one end, and the other end as its base. The volume of the cylinder, hemisphere and the cone are respectively in the ratio

(a) 2 : 3 : 2 (b) 3 : 2 : 1 (c) 3 : 1 : 2 (d) 1 : 2 : 3

- 68.** A conical cavity is drilled in a circular cylinder of height 15 cm and base radius 8 cm. The height and the base radius of the cone are also same. Then the whole surface area of the remaining solid is

(a) $440 \pi \text{ cm}^2$ (b) $240 \pi \text{ cm}^2$ (c) $640 \pi \text{ cm}^2$ (d) $960 \pi \text{ cm}^2$

- 69.** Smaller lead shots are to be prepared by using the material of a spherical lead shot of radius 1 cm. Same possibilities are listed in the statements given below

- I. The material is just sufficient to prepare 8 shots each of radius 0.5 cm.
II. A shot of radius 0.75 cm and a second shot of radius 0.8 cm can be prepared from the available material.

Which of the above statement is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 70.** Consider the following statements:

- I. The volume of a cuboid is the product of the lengths of its coterminal edges.
II. The surface area of a cuboid is twice the sum of the products of lengths of its coterminal edges taken two at a time.

Which of the above statement(s) is/are correct?

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 71.** Consider the following statements :

The length of a side of a cube is 1 cm. Which of the following can be the distance between any two vertices?

I. 1 cm II. $\sqrt{2}$ cm III. $\sqrt{3}$ cm

Select the correct answer using the code given below.

(a) Only I (b) Only II (c) Only III (d) All of these

- 72.** Consider the following statements.

- I. When eight drops of water combine to form a single drop, the surface area of all the eight drops is greater than the surface area of big drop.

- II. Square of volume of a spherical body is directly proportional to cube of its surface area.

- III. A sphere occupies the biggest space but has the smallest surface area.

Which of the above statement(s) is/are correct?

(a) I and II (b) I and III (c) II and III (d) All of these

- 73.** Consider the following statements:

- I. The curved surface area of a right circular cone of base radius r and height h is given by

$$\pi r(\sqrt{h^2 + r^2}).$$

- II. The right circular cone of base radius r and height h when cut opened along the slant height, forms a rectangle of length πr and breadth $\sqrt{h^2 + r^2}$.

- III. If a right circular cone is cut off by a plane parallel to the base of the cone, then its curved surface area is $\frac{\pi h}{3} [R^2 + r^2 + Rr]$.

- IV. If a right triangle is being revolved about one of its edges. The surface generated by the hypotenuse is the lateral surface of the right circular cone.

Which of the above statement(s) are correct?

(a) I and II (b) I, II and III (c) I, III and IV (d) All of these

Directions (Q. Nos. 74-76) Read the following information carefully and answer the given questions that follow.

A cone of height 10 cm and radius 5 cm is divided into two parts by drawing a plane through the mid-point of its axis, parallel to its base.

- 74.** What is the ratio of the volume of the original cone to the volume of the frustum left?

(a) $\frac{4}{3}$ (b) $\frac{7}{8}$ (c) $\frac{8}{7}$ (d) $\frac{9}{8}$

- 75.** What is the area of the top circle of the frustum in square centimetre?

(a) 25π (b) $\frac{5\pi}{4}$ (c) $\frac{25\pi}{4}$ (d) None of these

- 76.** What is the ratio of the curved surface area of the original cone and the curved surface area of the frustum?

(a) 3 : 1 (b) 3 : 2 (c) 4 : 1 (d) 4 : 3

Directions (Q. Nos. 77-79) Read the following information carefully and answer the given questions that follow.

Let C be a right circular cone. It is given that the two ends of a frustum of C are of radii 3 cm and 6 cm and the height of the frustum is 9 cm.

- 77.** What is the slant height of the given frustum?

(a) $3\sqrt{10}$ cm (b) $6\sqrt{10}$ cm (c) 12 cm (d) 15 cm

- 78.** What is the height of the cone?

(a) 9 cm (b) 12 cm (c) 13.5 cm (d) 18 cm